



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2018

BIOLOGY

9744/01

HIGHER 2

28 August 2018

Paper 1 Multiple Choice

1 hour

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, pencil clips, highlighters, glue or correction fluid.

Write your name, centre number and index number on the Answer Sheet provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

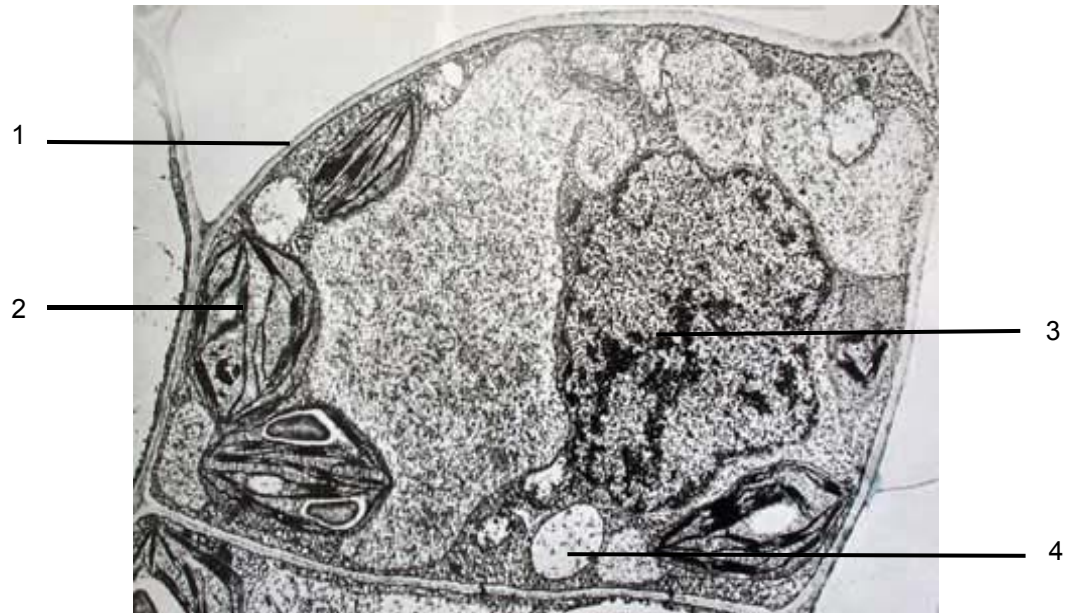
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

This question paper consists of **21** printed pages.

- 1 The electron micrographs show two different types of cells (not shown to scale).



Cell A



Cell B

Which row matches the structures to their correct function?

	Structure in cell A	Structure in cell B	Function
A	3	7	Provide energy for the cells
B	1	5	Maintain the shape of the cells
C	2	6	Photosynthesise
D	4	6	Secrete proteins

- 2** The plasma membrane is the cell's protective barrier as it prevents foreign molecules from entering the cell. However, in drug research experiments, foreign molecules such as drugs or short DNA fragments need to be transported into the cell.

Electroporation is a technique used to increase the permeability of the membrane transiently by treating the cell with short electrical pulses.

Which statement most likely explains how electroporation works?

- A** The short electrical pulses denature the membrane proteins, allowing foreign molecules to pass through.
- B** The short electrical pulses cause the foreign molecules to be attracted to the surface of the membrane.
- C** Electricity increases the hydrophobic nature of foreign molecules, allowing them to pass through the hydrophobic core of the phospholipid bilayer.
- D** Electroporation causes the phospholipids to move apart to create pores for foreign molecules to pass through.

- 3** Which row matches the descriptions of biological molecules to where they are found?

- 1 Polymer of glucose molecules linked by β -1,4 glycosidic bonds to form a straight chain
- 2 An unbranched and helical polymer of glucose molecules linked by α -1,4 glycosidic bonds
- 3 An amphipathic, phosphate-containing molecule

	1	2	3
A	Cell wall of eukaryotes only	Storage granules in animal cells	Plasma membrane of prokaryotes and eukaryotes
B	Cell wall of prokaryotes and eukaryotes	Storage granules in plant cells	Plasma membrane of eukaryotes only
C	Cell wall of prokaryotes and eukaryotes	Storage granules in animal cells	Plasma membrane of prokaryotes only
D	Cell wall of eukaryotes only	Storage granules in plant cells	Plasma membrane of prokaryotes and eukaryotes

- 4 Proteins can be categorised as globular or fibrous, depending on their structure.

How many of the following statements describes fibrous proteins only?

- 1 Monomers are linked by peptide bonds formed during translation
- 2 Long molecules which are insoluble in water
- 3 Helical molecules consisting of non-repeating sequences of amino acids
- 4 Contains hydrogen bonds between polypeptides

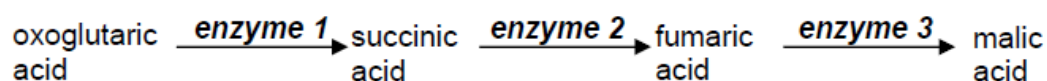
A 0

B 1

C 2

D 3

- 5 Malic acid is produced by a series of enzyme-catalysed reactions.



The addition of malonic acid results in an accumulation of succinic acid, a near absence of both fumaric acid and malic acid, and has no effect on the concentration of oxoglutaric acid.

Further addition of fumaric acid results in the formation of malic acid.

What does this information indicate about malonic acid?

- A It is an inhibitor of enzyme 1.
- B It catalyses the formation of succinic acid.
- C It is an inhibitor of enzyme 2.
- D It reacts with fumaric acid.

6 Which statement is correct regarding the enzymatic activity of catalase under the following conditions?

- 1 Addition of a non-competitive inhibitor
- 2 Addition of pH 13 buffer solution
- 3 Incubation at 10°C

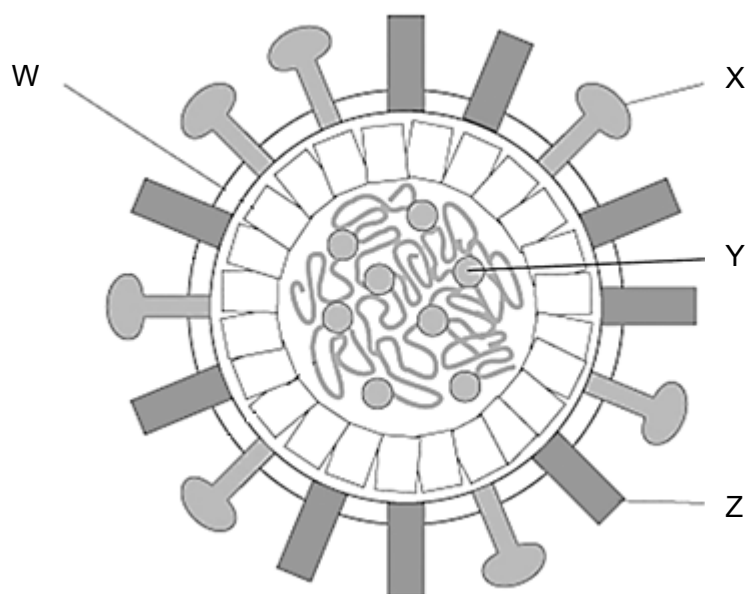
- A** The hydrogen and ionic bonds between R groups of residues are broken in conditions 2 and 3, hence the rate of reaction decreases.
- B** The catalytic and binding residues in all active sites are affected in condition 1, hence the rate of reaction decreases.
- C** The chances of effective collisions to form enzyme-substrate complexes are low in condition 3, hence the rate of reaction decreases.
- D** The changes in the 3D conformation of enzymes are irreversible in conditions 1, 2 and 3, hence the rate of reaction decreases.

7 Which statements regarding stem cells are true?

- 1 Researchers can induce embryonic stem cells to differentiate into various cells and tissue types to repair damaged tissue.
- 2 The use of embryonic stem cells for research can be an ethical challenge as the continued destruction of embryos could desensitise medical communities to the destruction of life.
- 3 One of the normal functions of blood stem cells in a living organism is the transplantation of such stem cells from normal healthy bone marrow donors to leukemia patients for treatment.
- 4 Blood stem cells can potentially differentiate into neurones under appropriate chemical signals.

- A** 1, 2 and 3 **B** 1, 2 and 4 **C** 1 and 3 **D** 2 and 4

- 8 The figure below shows a simplified representation of an influenza virus.



Which option does **not** describe the significance of the labelled structure?

	Structure	Significance
A	W	Acts as an additional protective coat obtained from host cell membrane via budding
B	X	Aids virus in its penetration of the mucus layer by breaking down mucoproteins
C	Y	Allows synthesis of viral genome and proteins
D	Z	Binds to sialic acid-containing receptors on epithelial cells and cleaves them for entry

- 9** Binary fission allows the bacteria to divide and gives rise to genetically identical daughter cells. Scientists are interested in uncovering the genetic mechanisms that regulate and drive binary fission, so as to design novel antibiotics that could specifically target and interfere with the process.

They found out that during binary fission, bacteria produce the protein FtsZ which assemble into a ring-like structure at the centre of the cell. This causes the cytoplasm to divide into two, before synthesis of a new wall begins.

Which of the following statements can be deduced from the description above?

- 1 The genes that code for FtsZ can be found within a plasmid.
- 2 An antibiotic that inhibits the action of FtsZ will result in only one cell with more than one chromosomes.
- 3 Mutations in genes coding for FtsZ will prevent the replication of DNA in binary fission.
- 4 The replicated chromosomes attach at the opposite ends of the parent bacterial cell before it divides into two daughter cells.

A 1, 2, 3, and 4 **B** 1 and 3 only **C** 2 and 4 only **D** 2 only

- 10** Which statement(s) about DNA polymerases and RNA polymerases is/ are correct?

- 1 They read the DNA template in the 3' to 5' direction.
- 2 They unwind and unzip double-stranded DNA.
- 3 They read the terminator sequence and stop adding nucleotides to nucleic acid chains.
- 4 They bind to the same specific sequences to start their processes.

A 1 and 3 **B** 2 and 3 **C** 1 only **D** 4 only

11 The following steps describe a method to study the structure and localisation of protein kinase c in yeast cells.

- tRNAs with anticodon AAG were isolated and chemically modified to carry their specific fluorescent amino acids.
- During translation, chemically modified amino-acyl tRNAs bind to the large ribosomal subunit.
- Fluorescent amino acids are incorporated into the elongating polypeptide chain.
- After translation, the polypeptide chain folds into its native conformation and the positions of the fluorescent amino acids can be detected and studied.

Which of the following describes a condition which will allow the above method to be carried out?

- A** Peptidyl transferase is specific to the fluorescent amino acid and the elongating polypeptide chain in order to catalyse the formation of a peptide bond.
- B** Synthetic amino-acyl tRNA synthetase is specific to the fluorescent amino acid and chemically modified tRNA in order to form modified amino-acyl tRNA.
- C** The ribosome is specific to the mRNA sequence in order to synthesise protein kinase C.
- D** Chemically modified amino-acyl tRNA is specific to the P site of a large ribosomal subunit in order to add the fluorescent amino acid to the elongating polypeptide chain.

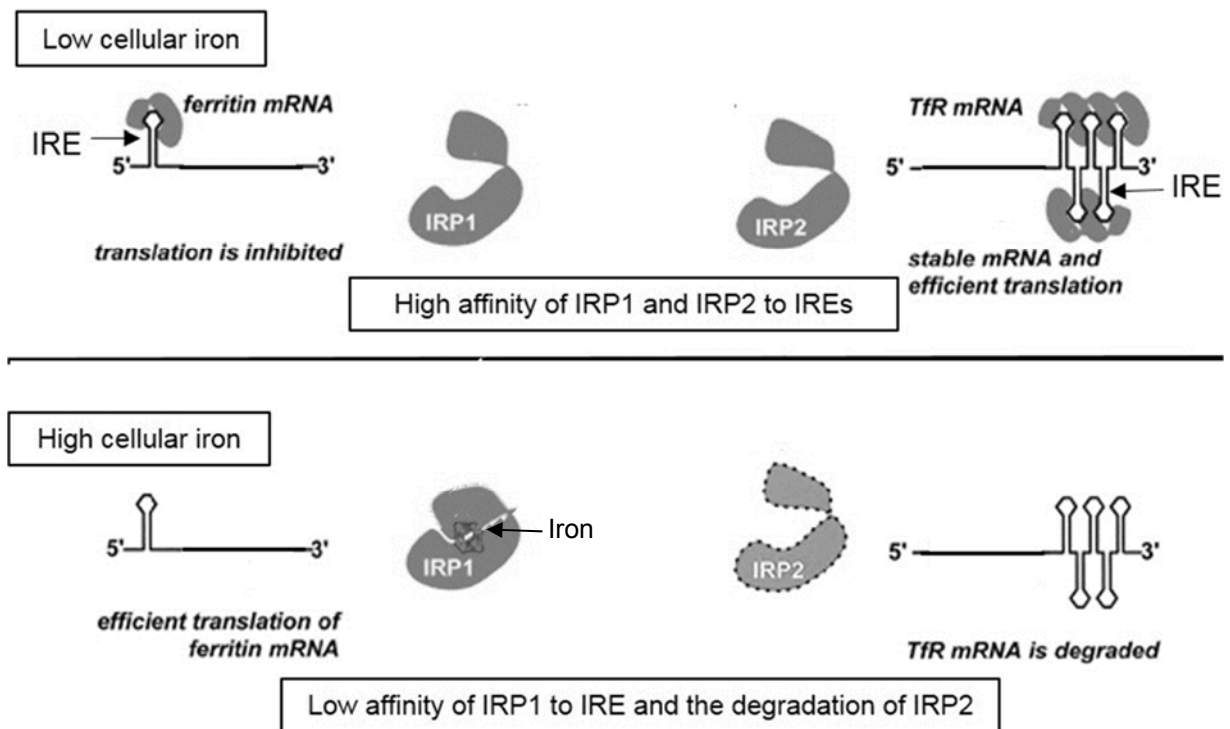
12 Which of the following statement(s) regarding the polymerase chain reaction (PCR) is/ are **not** true?

- 1 Without knowing the target sequence to be amplified, PCR cannot be carried out because primers cannot be synthesized.
- 2 There is a high fidelity of target sequence amplification due to the proof-reading activity of *Taq* polymerase.
- 3 The amplification of the target sequence can still continue even when the starting PCR mixture is contaminated with DNA molecules from another origin.
- 4 If four copies of the target sequence are present after the first PCR cycle, there will be 2^{201} copies of the target sequence after 200 cycles.

- A** 1 and 4 **B** 2 and 3 **C** 2 and 4 **D** 2 only

- 13** In mammalian tissues, iron enter cells via transferrin receptors (TfR) and are stored when they are bound by ferritin proteins. The synthesis of TfR and ferritin is regulated through translational controls by regulatory proteins IRP1 and IRP2, which bind to iron-responsive elements (IREs) found on mRNAs.

IREs are loops found at the 5' UTR of ferritin mRNA and 3' UTR of TfR mRNA respectively. The effects of different iron levels on IRP1 and IRP2 are shown in the figure.



Which row correctly describes the translational control of ferritin mRNA and TfR mRNA?

	Cellular iron (Fe) levels	Ferritin levels	TfR levels
A	High	High	High
B	Low	Low	Low
C	High	High	Low
D	Low	High	Low

- 14 The table shows the genome size, number of genes and chromosome number of the following organisms:

Organism	Genome size (bp)	Number of genes	Chromosome number
<i>E. coli</i> bacterium	4 369 000	4289	$n = 1$
Baker's yeast	12 069 000	6200	$2n = 32$
Amoeba	290 000 000 000	Data not available	500 – 1000 (possibly polyploidy)
Rodent	3 399 900 000	Data not available	$2n = 64$
Humans	3 200 000 000	30 000	$2n = 46$

From the data provided, it is possible to conclude that

- 1 The less complex the organism, the smaller the genome size.
- 2 Amoeba has more non-coding DNA than humans.
- 3 The larger the genome size, the higher the number of chromosomes.
- 4 The number of housekeeping genes is the same for all organisms.

A 1 and 2

B 2 and 4

C 3 only

D 4 only

- 15 The arginine (*arg*) operon is present in some bacteria to regulate the synthesis of the essential amino acid arginine. The operon consists of structural genes that code for the enzymes that synthesise arginine. The operon is switched off when arginine is in excess.

Which of the following statements is **not** true?

- A** Excess arginine acts as the co-repressor which binds to the repressor.
- B** Excess arginine acts as the repressor which binds to the operator.
- C** The *arg* operon is repressible.
- D** The regulation of *arg* operon is similar to that of the *trp* operon.

- 16** A species of lizards, Whiptail lizards, reproduce only by parthenogenesis. This mode of asexual reproduction allows females to produce offspring alone, without the genetic contribution of a male.

Scientists are worried that this species may become endangered as a result of climate change.

Which statements support the scientists' concerns?

- 1 Offspring of parthenogenetic species are genetically identical.
- 2 Parthenogenesis produces too few offspring for a viable population.
- 3 Genetic contribution of a male is required in order for the offspring to survive.
- 4 In asexual reproduction, meiosis does not occur to produce genetically different gametes.

A 1, 2, 3 and 4 **B** 1, 2 and 4 **C** 1 and 4 only **D** 2 and 3 only

- 17** Elephants have been found to be strangely resistant, though not immune, to cancer. They are four times less likely to develop cancer as compared to humans. Elephants have 40 copies of the *p53* gene while humans only have two. In addition, three genes which code for DNA repair proteins have found to be very active.

What do these observations suggest?

- A** All DNA mutations in the elephant are repaired.
- B** When elephants develop cancer, it is due to environmental causative factors such as excessive exposure to UV light.
- C** The occurrence of cancer is due to the *p53* oncogene, which stimulates the cell to divide rapidly, bypassing the cell cycle checkpoints.
- D** There is large amount of *p53* proteins in an elephant cell which prevent cells with mutations from moving past the cell cycle checkpoints.

- 18 Rabbits carry the *C* gene which is required for the development of pigments in their fur. The alleles show a hierarchy of dominance when present in heterozygous individuals as shown in the figure. The order of dominance of these alleles, in descending order, is *C*, *c^{ch}*, *c^h*, *c*.

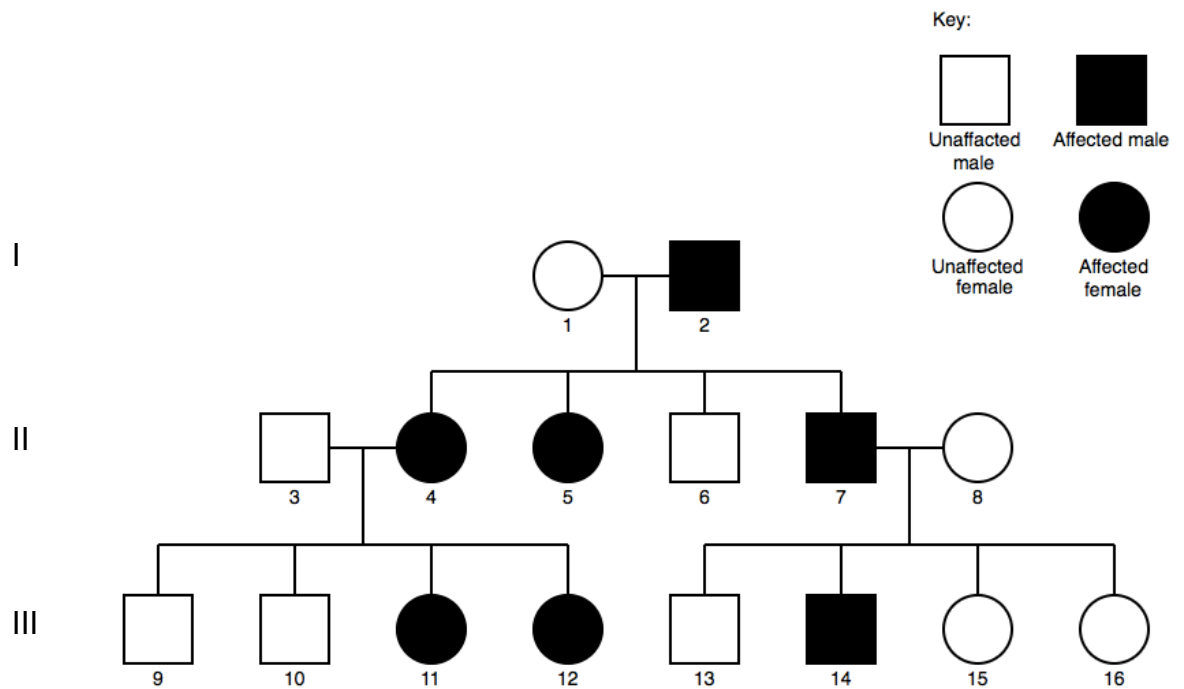
Allele			
<i>C</i>	<i>c^{ch}</i>	<i>c^h</i>	<i>c</i>
Genotype			
<i>CC</i>	<i>c^{ch}c^{ch}</i>	<i>c^hc^h</i>	<i>cc</i>
Phenotype			
WILD TYPE: Brown fur	CHINCHILLA: Black-tipped white fur	HIMALAYAN: White fur with black paws, nose, ears, tail	ALBINO: White fur
			

A scientist observed that the fur on the paws, noses and ears of Himalayan rabbits tends to be black while the rest of its body tends to be white. The scientist extracted a section of skin from the ear of a Himalayan rabbit and cultured the follicle cells. He discovered that when the cells are exposed to temperatures between 15°C to 25°C, they synthesise certain pigments. However, at temperatures beyond 35°C, these pigments are not synthesised.

Which statement best explains the results of the scientist's experiments?

- A Temperatures beyond 35°C alter the structure of the pigments produced by the skin cells, hence the rest of the Himalayan rabbit's body tends to be white.
- B The Himalayan phenotype is a result of the *c^h* allele that produces a temperature-sensitive gene product which controls the production of the pigments.
- C Multiple alleles of the *C* gene can give rise to white fur with black patches or white fur under different temperatures.
- D The Himalayan phenotype is an example of incomplete dominance where it is an intermediate of the chinchilla (*c^{ch}c^{ch}*) and albino (*cc*) genotypes.

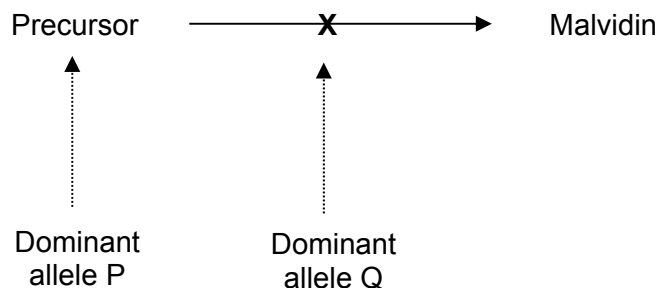
- 19 The figure shows the pedigree for the inheritance of polycystic kidney disease.



Assuming that the letters R and r denote dominant and recessive alleles respectively, what is the mode of inheritance for this disease and the possible genotype of individual 11?

	Mode of inheritance	Genotype of individual 11
A	Autosomal dominant	RR
B	Autosomal recessive	rr
C	X-linked dominant	$X^R X^r$
D	X-linked recessive	$X^r X^r$

- 20 In the *Primula* plant, production of the pigment malvidin creates blue-coloured flowers. Synthesis of malvidin is controlled by dominant allele P at the P/p gene locus which codes for a precursor substance. However, the conversion of the precursor substance to malvidin can be suppressed by the dominant allele Q at the gene locus Q/q as shown in the biochemical pathway. Flowers with the absence of malvidin will appear white.



When two plants with white flowers are crossed, the F₁ generation has 44 plants with white flowers and 10 plants with blue flowers.

Which of the following statements is **not** true?

- A The probability of obtaining a pure-breeding offspring at both loci with white flowers is 1 of 8.
 - B The deviation from the Mendelian ratio (9:3:3:1) is due to gene interaction in which gene Q alters the phenotypic expression of gene P.
 - C The genotypes of the parents must be PpQq x PpQq.
 - D The genotypes PPQQ, PPQq, PpQq and ppQq produce the same phenotype.
- 21 Through a series of energy transfers, chemiosmosis results in the generation of ATP during aerobic respiration and photosynthesis.

Which of the following statements describe the conditions necessary for the chemiosmotic synthesis of ATP in both processes?

- 1 The transport of a lipid-soluble molecule against its concentration gradient.
- 2 The presence of oxygen as a final electron and proton acceptor.
- 3 The impermeability of the phospholipid bilayer to substances which are polar or charged.
- 4 The transmembrane nature of electron carrier proteins.
- 5 The provision of a pore which allows the facilitated diffusion of molecules.

- A 1, 3 and 5 B 2, 4 and 5 C 2, 3 and 4 D 3, 4 and 5

22 The electronmicrograph shows structures found in a cell.



Which row matches the events occurring at the labelled structures?

	X	Y	Z
A	$\text{ADP} + \text{P}_i \rightarrow \text{ATP}$	$\text{ADP} + \text{P}_i \rightarrow \text{ATP}$	$\text{NADPH} \rightarrow \text{NADP}^+ + \text{H}^+ + 2\text{e}^-$
B	$\text{ATP} \rightarrow \text{ADP} + \text{P}_i$	$\text{ATP} \rightarrow \text{ADP} + \text{P}_i$	$\text{NADP}^+ + \text{H}^+ + 2\text{e}^- \rightarrow \text{NADPH}$
C	$\text{NAD}^+ + \text{H}^+ + 2\text{e}^- \rightarrow \text{NADH}$	$\text{H}_2\text{O} \rightarrow 2\text{H}^+ + 2\text{e}^- + \frac{1}{2}\text{O}_2$	$\text{CO}_2 + \text{RuBP} \rightarrow 2\text{PGA}$
D	$\text{FAD} + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{FADH}_2$	$2\text{H}^+ + 2\text{e}^- + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$	$\text{Acetyl CoA} + \text{Oxaloacetate} \rightarrow \text{Citrate}$

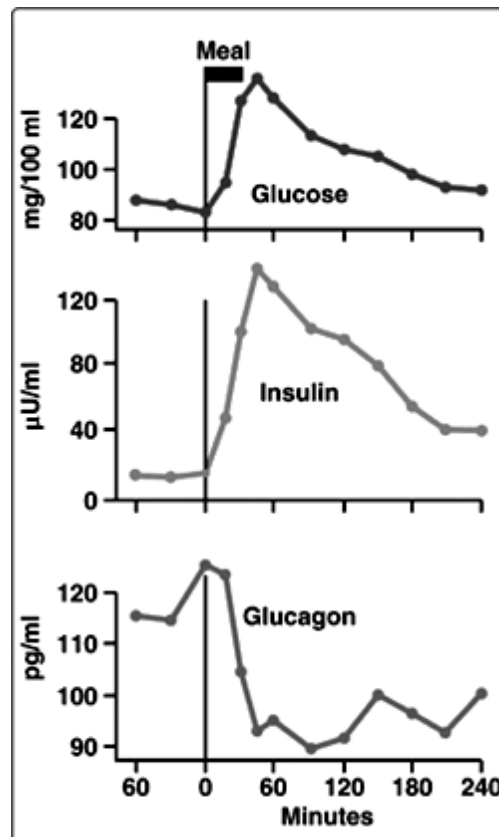
- 23** Nerve cells in the brain communicate by chemical molecules known as neurotransmitters. The following events, in the order shown, describe how binding of a neurotransmitter to a nerve cell will elicit a cellular response.

- 1 Neurotransmitter binds to a G-protein linked receptor (GPLR) at the cell surface
- ↓
- 2 A change in conformation of the GPLR allows the binding of a G protein to its intracellular domain
- ↓
- 3 G protein releases a GDP molecule in exchange for GTP
- ↓
- 4 The alpha subunit of the G protein dissociates from the other subunits
- ↓
- 5 The alpha subunit of the G protein activates adenylyl cyclase, which in turn converts ATP into cyclic AMP (cAMP)
- ↓
- 6 cAMP activates protein kinase A (PKA), which in turn phosphorylates transmembrane protein channels
- ↓
- 7 Phosphorylated protein channels open, allowing the facilitated diffusion of specific ions into the nerve cell

How many of the events allow for the amplification of signal to occur?

- A** 3
- B** 4
- C** 5
- D** 6

- 24 The figure shows the levels of glucose, insulin and glucagon found in blood, before and after a carbohydrate-containing meal was ingested.



Which row correctly identifies the events occurring at the respective timings?

	Timing	Events
A	60 minutes before meal	Glucagon results in the activation of glycogen phosphorylase, leading to increased rates of glycogenolysis in liver cells and an increase in blood glucose levels.
B	30 minutes before meal	Glucagon binds to G protein-linked receptors, leading to the activation of G proteins due to the hydrolysis of GTP molecules. Cellular responses lead to the maintenance of blood glucose levels.
C	At the start of meal (0 minutes)	Insulin binds to tyrosine kinase receptors, resulting in the cross-phosphorylation of receptor monomers. Cellular responses lead to a decrease in blood glucose levels.
D	60 minutes after meal	Insulin results in the translocation of glucose transporters to the cell surface, leading to increased rates of glucose uptake in muscle cells and a decrease in blood glucose levels.

- 25** Scientists examined the evidences used to establish the relationships between whales and other mammals such as hippopotamuses, pigs and cows.

The DNA sequences of common genes were examined in mammals, such as the haemoglobin beta-globin gene and the beta-casein gene. The beta-casein gene codes for beta-casein protein which is secreted in milk. The number of nucleotide differences in the beta-casein gene for four mammals is shown in the table.

	Whale	Hippopotamus	Cow
Pig	11	11	13
Cow	9	8	
Hippopotamus	3		

In addition, the fossil records showed that the most recent common ancestors of whales lived in the aquatic environment between India and Asia and their diet consisted mainly of fish or meat. However, the most recent common ancestors of hippopotamuses lived along the swampy coastal areas on the Asia continent and mainly fed on plants.

Which of the following deductions can be made from the evidences that scientists have gathered?

- A** Based on the molecular data, whales and pigs are more closely related than whales and hippopotamuses.
- B** Divergent evolution of the whales and hippopotamuses occurred due to limited gene flow between the ancestors.
- C** As there are no homologies present in the molecular data, whales are not related to the pigs and cows.
- D** Hippopotamus diverged from a common ancestor more recently than pigs, cows and whales.

- 26** Due to various human activities, there are many species that have become endangered. As these species face a reduction in population size, they are at risk of becoming extinct. It is found that the endangered species lose their genetic variation as the population size is reduced. Should the endangered population size increases, the genetic variation will not increase much within the next hundred years.

Which of the following statements is **not** true?

- A** A population would need to accumulate many heritable mutations over many generations in order to increase its genetic variation.
 - B** As the population size of the endangered species is reduced, there will be a loss of alleles due to the death of the individual organisms.
 - C** Epidemics could kill the endangered species easily as the population has low genetic variation, increasing their chance of extinction.
 - D** As the population size is reduced, sexual reproduction with random mating within the endangered species will increase heterozygosity, resulting in higher relative fitness in the population.
- 27** A population of green anole lizards (*Anolis carolinensis*) is native to the trees in Florida in the United States of America (USA). They tend to occupy the lower branches of the trees which are thicker, instead of the higher branches which are thinner and snap more easily.

In the 1950s, the brown anole lizards (*Anolis porcatus*) from Cuba were introduced into Florida. These lizards are larger and heavier than the green anole lizards and also prefer the lower branches. Both the green and brown anole lizards feed on flying insects that fly around the branches. The adults of both species also tend to feed on the hatchlings of the other species.

Studies have shown that after 20 years, the green anole lizards have occupied the higher branches. Their toe pads are also larger than before, with sticky scales so that they can perch on the higher branches.

Which statement best explains how the evolution of the green anole lizard in Florida has occurred?

- A** Mating between the green and brown anole lizards during the 20 years results in the green anole lizards having larger toe pads and sticky scales, on which natural selection could act upon.
- B** Within the original green anole lizard population, there were some individuals that already had larger toe pads and sticky scales, on which natural selection could act upon.
- C** Green anole lizards climbed onto higher branches as the brown anole lizards could not do so.
- D** Green anole lizards which live longer usually leave more offspring since they have more reproductive opportunities.

- 28** The Varivax® vaccine contains attenuated varicella zoster virus for the immunisation and prevention of chickenpox.

The virus was propagated in human diploid cell cultures, where trace quantities of an antibiotic known as neomycin were introduced to prevent bacterial contamination during the manufacturing process.

The first dose of the vaccine is administered by injection into the fatty layer under the skin, and a booster dose is given at least 3 months after the first dose.

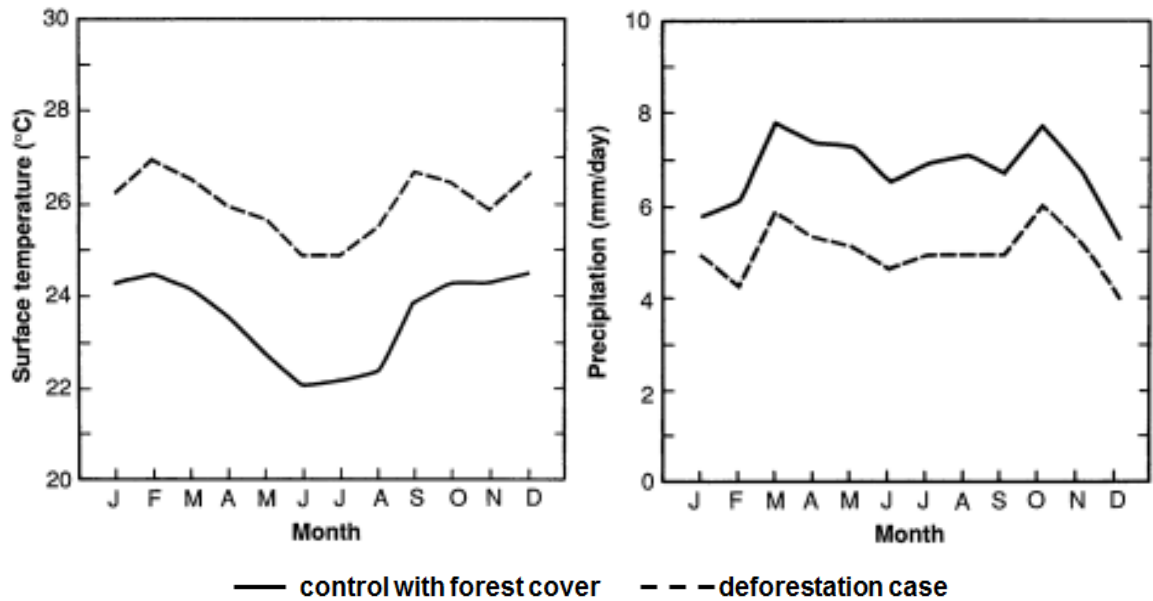
Which of the following statements about the Varivax® vaccine is **not** true?

- A** A booster dose is required because it is possible for memory cells to decline over time, causing immunity to fall below protective levels.
 - B** The vaccine should not be used on a patient who is immunodeficient.
 - C** The vaccine confers natural, active immunity in individuals who are injected with it.
 - D** Allergic reactions in the form of rashes may be expected in some individuals who are given the vaccine.
- 29** During an infection by the dengue virus (DENV), the pathogen which causes dengue fever, the secretion of antibodies by plasma cells forms part of the immune response mounted by the body.

Which option correctly explains why antibodies may or may not be effective against DENV?

	Effective against DENV	Not effective against DENV
A	Antibodies bind to extracellular DENV, preventing uptake by host cell through receptor-mediated endocytosis.	The extracellular phase constitutes a large part of the DENV reproductive cycle.
B	Antibodies bind to DENV, triggering the release of histamine by mast cells which results in an inflammatory response.	The RNA genome of DENV is prone to mutations, hence antibodies produced may not be able to bind to new viral strains.
C	Antibodies neutralise toxins secreted by viruses by preventing the binding of toxins to host cells.	DENV may undergo a latent phase where no viral proteins are synthesised and secreted.
D	Antibodies bind to extracellular DENV, facilitating their uptake by phagocytes.	Antibody-mediated opsonisation may facilitate the infection of macrophages which are target cells of DENV.

- 30** Scientists created a simulation of the effects of Amazon deforestation on the climate. In the simulation, an area of the tropical forest was replaced by a pasture of grass. The simulation provided monthly data on the temperature of the ground surface (surface temperature) and rainfall (precipitation) over a year-long period.



Which statements could explain these data?

- 1 Grass is a more effective carbon sink than trees.
- 2 The removal of forest cover disturbed the soil layers and resulted in carbon emissions.
- 3 Without the trees, the reduced transpiration rate resulted in less water vapour released into the atmosphere.
- 4 Increase in surface temperature could lead to increase in the rate of evaporation.

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 2 and 4